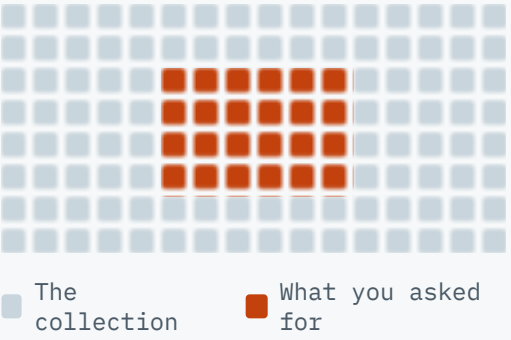
Read the part you need.

BIL is one interface to imaging sensor data in any format. It extracts the subset an algorithm needs, as native pixels or as 2D float32 arrays, and leaves the rest unread.



1.8
GIGAPIXELS

One frame from the ARGUS-IS sensor is 7.2 GB once held as float32. Nobody processes it whole. BIL is designed so that nobody has to load it whole.

PROPORTIONAL

Cost follows the request

A small window of an enormous file costs about what it costs from a small one.

UNIFYING

One interface, many formats

NITF, GeoTIFF, JPEG 2000, SICD, ENVI, HDF5, raw camera files and video.

READY TO PROCESS

Arrays your algorithms take

Named float32 Slices at any resolution level, zero-copy where the storage allows.

CONTAINED

Safe on closed networks

No network access unless you ask. Security markings travel with everything written.

```
image = bil.open_source("scene.ntf").image(0)
slice = image.read_slice("complex.polar.magnitude", window)
```

In development. The first release reads GeoTIFF and NITF.

Rust core · Python, C and C++ interfaces · LGPL 2.1 or later

READ THE SPECIFICATION

github.com/code-moth/BlazingImagingLibrary

Issues and discussion on GitHub

From file to array.

1 OPEN

Source

One file, a set of files or a stream.

2 READ

Fragment

The native pixels of a window, zero-copy where storage allows.

3 PROCESS

Slice

A named 2D float32 array, such as magnitude or hue.

FIVE WAYS TO CUT THE DATA DOWN

Too much spatial extent	**Window**	Take a rectangle
Too much resolution	**Level**	Take a coarser version
Too many bands	**Slice**	Take one 2D view
Too much time	**Frame**	Take one frame
Too many sample types	**float32**	One type for every sensor

LEVELS

Any resolution, built in

Filtered levels, fast decimation for overviews, and the file's own stored levels by name.

GEOCODING

Locates every pixel

One chain of steps covers map grids, RPC, camera and SAR models, in both directions.

METADATA

Fields on demand

Named fields under their native names, extracted only when you ask for them.

PLUGINS

Extends to new formats

Add drivers in Rust, C or C++, or script them in Python while the core reads the pixels.

DESIGNED TO READ

Optical and infrared stills, SAR, multispectral and hyperspectral cubes, raw camera files, video and wide area motion imagery, LIDAR and medical volumes.

First release: GeoTIFF and NITF.

Next: ENVI, video with KLV, SICD and SIDD.

Rust core · Python, C and C++ interfaces · LGPL 2.1 or later

READ THE SPECIFICATION

github.com/code-moth/BlazingImagingLibrary

Issues and discussion on GitHub